HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed technical delivery of geotechnical projects, developing schedules and coordinating resources to meet program and budget objectives for a national infrastructure program.
- Provided consultative advice to external partners and clients on site selection, optioneering, and design stages for large-scale civil engineering projects.
- Oversaw peer review and verification of geotechnical data and design calculations, ensuring adherence to current design codes and best practices for complex civil works.
- Mentored and guided junior members of the Ground Engineering team, providing technical support and development opportunities to foster their growth.
- Collaborated with Transport, Water, and Energy teams to provide geotechnical inputs for multi-disciplinary project tenders and delivery, ensuring integrated solutions.
- Maintained and developed relationships with key clients and prospective customers, identifying new business opportunities and promoting ground engineering services.
- Deployed containerized Java Spring Boot services to AWS EC2 instances using Docker, configuring basic resource limits and monitoring via CloudWatch for production stability.

SKILLS

Geotechnical Engineering – Geotechnical data analysis, Design calculations, Ground investigation, Soil mechanics, Geotechnical design

Project Management – Project planning, Scheduling, Budget management, Client relationship management, Stakeholder management

Software Development – Java, Spring Boot, REST APIs, Hibernate, JPA, JUnit, Docker, AWS (EC2, CloudWatch)

Cloud & DevOps – AWS, Docker, Jenkins, Git, Linux

Data Analysis – SQL, PostgreSQL, MySQL, Oracle, Power BI, Tableau, R, Python (pandas, NumPy, scikit-learn)

PROJECTS

Enterprise Geotechnical Data Management System

- Developed a Java Spring Boot application with a PostgreSQL backend to manage and analyze geotechnical investigation data for a large infrastructure client.
- Implemented REST APIs for data ingestion and retrieval, integrating with existing client systems for seamless data flow and reporting.
- Designed data models using JPA and Hibernate to represent complex soil strata and test results, ensuring data integrity and efficient querying.

Automated Site Assessment Tool

- Created a Python script that processes geological survey data, performing parametric assessments and correlations for common geotechnical design problems.
- Integrated the script with a Power BI dashboard to visualize potential site risks and provide preliminary optioneering recommendations to engineering teams.
- Designed and implemented a data pipeline using pandas and NumPy to clean and transform raw survey outputs into actionable insights for project planning.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)